



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: IX
SUBJECT: MATHEMATICS

FULLMARKS: 40
DATE: 18.06.24

General Instructions:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. Choose the correct option

[1 × 5 = 5]

i) If $a + b = 10$ and $ab = 21$ then, the value of $a^2 + b^2$ is:

- a) 37 b) 58 c) 42 d) 100

ii) Among the given numbers the irrational number is:

- a) $\sqrt{23}$ b) $\sqrt{225}$ c) 0.3796 d) 7.478478....

iii) The term to be added to $9x^2 + 36y^2 + 10xy$ to make it a perfect square trinomial is:

- a) $4xy$ b) $16xy$ c) $-12xy$ d) $26xy$

iv) The value of $8^3 - 20^3 + 12^3$ is:

- a) 5550 b) 7650 c) - 5760 d) - 5070

v) Assertion (A): Compound interest is equal to simple interest for the period of 1 year but it is always less than simple interest for more than 1 year. (assume interest is calculated yearly on the same amount with same rate of interest in both CI and SI).

Reason (R): The compound interest for an amount depends on both Principal and interest gained over periods.

- a) (A) is true but (R) is false b) (A) is false but (R) is true
c) Both (A) and (R) are true d) Both (A) and (R) are false

SECTION: B

2. Factorise:

[3+3]

i) $35y^2 + 13y - 12$

ii) $a(a+b)^3 - 4a^2b(a+b)$

3. By using suitable algebraic identities, evaluate the following: $(110)^3$ [3]

4. At what rate per cent compound interest, does a sum of money become 1.44 times of itself in 2 years? (interest compounded annually) [3]

5. Insert three rational numbers between $\frac{1}{12}$ and $\frac{3}{5}$. [3]

6. A principal of ₹6400 is kept in a bank and the rate of interest is 5% p.a compounded half-yearly.

i) Calculate the amount when the duration is 1 year.

ii) What will be the amount after $1\frac{1}{2}$ years?

iii) Calculate the interest after $1\frac{1}{2}$ years. [5]

7. If $x + \frac{1}{x} = 6$, then find the value of a) $x^2 + \frac{1}{x^2}$ b) $x^4 + \frac{1}{x^4}$ [5]

8. Evaluate: $\frac{4}{1+2\sqrt{3}} - \frac{6+\sqrt{3}}{11\sqrt{3}}$ [5]

9. Present value of an article is ₹45000. The value of the article decreases for the first two years at the rate of 10% per year and then in the third year it increases by 10%. Find the value of the article after 3 years. By how much has the value increased or decreased? [5]